STRUCTURE MAPS OF A REDUCED GROUPOID OVER A PERFECT FIELD OF CHARACTERISTIC P ARE SMOOTH

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The sheaf of differentials of R to U for a groupoid over a field with s, t locally of finite type is free (more-groupoids chapter, Lemma lemma-sheaf-differentials, canonical tag 04R9; collected on this lane as L22526). cited at proof lines 1652-1654. Provides the freeness of the sheaf of differentials Omega_{R/U}, the input to the smoothness criterion.
- Over a perfect field a locally finite type morphism from a reduced scheme with free sheaf of differentials is smooth (Varieties chapter, Lemma varieties-lemma-char-p-differentials-free-smooth, canonical tag 04QP). cited at proof lines 1654-1655. Turns freeness of the differentials, reducedness of R and local finite type into smoothness of s and t.

Source reference. Original lemma 04R9: sources/stacks-chow/tags.txt, source lines 6177–6177. Cited at proof lines 1652-1654: the sheaf Omega_{R/U} is free.

Source reference. Original lemma 04QP: sources/stacks-chow/tags.txt, source lines 6157–6157. Cited at proof lines 1654-1655: over a perfect field, free differentials plus reduced source plus locally finite type give smoothness.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume*

(1) *$U = \mathrm{Spec}(k)$ with k a field,*
(2) *s, t are locally of finite type,*
(3) *R is reduced, and*
(4) *k is perfect.*

Then $s, t : R \to U$ are smooth.

Proof. By Lemma 04R9 the sheaf $\Omega_{R/U}$ is free. Hence the lemma follows from Varieties, Lemma 04QP. □

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